Rv1255c, a dormancy-related transcriptional regulator of TetR family in Mycobacterium tuberculosis enhances isoniazid tolerance in Mycobacterium smegmatis

Article *in* The Journal of Antibiotics · September 2023

DOI: 10.1038/s41429-023-00661-8

The Journal of Antibiotics (2023) 76:720–727
https://doi.org/10.1038/s41429-023-00661-8

Japan Antibiotics Research Association
The Society for Actinomycetes Japan

ARTICLE

Rv1255c, a dormancy-related transcriptional regulator of TetR family in *Mycobacterium tuberculosis*, enhances isoniazid tolerance in *Mycobacterium smegmatis*

Jijimole Gopi Reji[1] · Lakshmi K. Edison[1] · Sajith Raghunandanan[1] · Akhil Raj Pushparajan[1] · Krishna Kurthkoti[1] · Ramakrsihnan Ajay Kumar[1]

Received: 16 February 2023 / Revised: 29 August 2023 / Accepted: 12 September 2023 / Published online: 11 October 2023

Abstract

Mycobacterium tuberculosis is exposed to diverse stresses inside the host during dormancy. Meanwhile, many metabolic and transcriptional regulatory changes occur, resulting in physiological modifications that help *M. tuberculosis* to adapt to these stresses. The same physiological changes also cause antibiotic tolerance in dormant *M. tuberculosis*. However, the transcriptional regulatory mechanism of antibiotic tolerance during dormancy remains unclear. Here, we showed that the expression of *Rv1255c*, an uncharacterised member of the tetracycline repressor family of transcriptional regulators, is upregulated during different stresses and hypoxia-induced dormancy. Antibiotic tolerance and efflux activities of *Mycobacterium smegmatis* constitutively expressing *Rv1255c* were analysed, and interestingly, it showed increased isoniazid tolerance and efflux activity. The intrabacterial isoniazid concentrations were found to be low in *M. smegmatis* expressing *Rv1255c*. Moreover, orthologs of the *M. tuberculosis katG*, gene of the enzyme which activates the first-line prodrug isoniazid, are overexpressed in this strain. Structural analysis of isoforms of KatG enzymes in *M. smegmatis* identified major amino acid substitutions associated with isoniazid resistance. Thus, we showed that *Rv1255c* helps *M. smegmatis* tolerate isoniazid by orchestrating drug efflux machinery. In addition, we showed that *Rv1255c* also causes overexpression of *katG* isoform in *M. smegmatis* which has amino acid substitutions as found in isoniazid-resistant *katG* in *M. tuberculosis*.

Introduction

Tuberculosis (TB) has always been one of the leading causes of morbidity and mortality from an infectious agent [1]. The recent COVID-19 pandemic has aggravated it with increased indoor exposure and decreased interventions [2]. In conjunction with an increased global report of phenotypic drug tolerance, multidrug-resistant TB (MDR-TB, resistant to not less than two of the potent first-line drugs, isoniazid and rifampicin) and extensively drug-resistant TB (XDR-TB, MDR-TB resistant to fluoroquinolones and at least one of the injectable second-line drugs), the forecast is much worse for the coming few years [3]. Thus, it is necessary to understand the mechanisms for drug resistance in *Mycobacterium tuberculosis* to remove the major obstacles to successful TB management.

Within the host, the intracellular *M. tuberculosis* is exposed to an arsenal of antibacterial mechanisms in macrophages and granuloma, such as hypoxia, reactive oxygen species (ROS), nitric oxide (NO), nutrient starvation, acidic pH, etc. [4]. The bacteria adapt to the stresses with physiological changes that allow them to tolerate these stresses and persist [5]. Studies suggest that many transcriptional and metabolic regulators are differentially expressed during stress and dormancy. For instance, exposure to hypoxic conditions has multifaceted effects on the bacteria, including replication arrest, shifts in lipid composition, metabolism and respiration [6, 7]. Various studies have established that isoniazid kills over 95% of actively growing *M. tuberculosis* bacilli. However, during hypoxic conditions,

Supplementary information The online version contains supplementary material available at https://doi.org/10.1038/s41429-023-00661-8.

✉ Ramakrsihnan Ajay Kumar
rakumar@rgcb.res.in

[1] Mycobacterium Research Laboratory, Rajiv Gandhi Centre for Biotechnology, Thiruvananthapuram 695014, India

the effect of isoniazid on bacilli is relatively low, with over 50% survival [8]. Furthermore, Ojha et al. demonstrated that a high number of *M. tuberculosis* cells survived isoniazid treatment in low-oxygen biofilms [9]. Flentie et al. reported that a small molecule, C10, can block the isoniazid tolerance caused by hypoxic stress [10]. Stresses, such as hypoxia, NO, and nutrient starvation, are known to induce dormancy in *M. tuberculosis* and make it resistant to drugs [4, 11]. Although physiological changes to dormancy and stresses can induce the formation of drug-tolerant *M. tuberculosis* [12, 13], the molecular and regulatory mechanisms that lead to this antibiotic tolerance still remain obscure. In addition, it has been shown that efflux pumps cause antibacterial resistance [14–17], yet the mechanisms of drug efflux during dormancy and stress are unclear [18].

Rv1255c is an uncharacterised tetracycline repressor family of transcriptional regulators (TFTR) and the gene is present in the RD10 (region of difference) in *M. tuberculosis* [19]. *Rv1255c* is a highly immunogenic antigen, and a promising biomarker for TB diagnosis [20]. Rustad et al. have shown that *Rv1255c* is induced temporally after the peak expression of *dosR* regulon, and the sensor kinase *dosS* in *M. tuberculosis* under hypoxia [21]. In addition, it is also reported that *Rv1255c* is a late hypoxic response gene downstream of the metabolic regulator *Rv3334* [22], which in turn is induced by the DosS/DosT-DosR regulon [11]. Galagan et al. have recorded in their ChIPseq data that Rv1255c also binds to the promoter of many target genes, including i*nhA*, during hypoxia-induced dormancy [23]. However, the role of *Rv1255c* in INH tolerance is unknown. Here we have shown that *Rv1255c* expression is elevated during various stresses and hypoxia-induced dormancy in vitro. We tested *Rv1255c* for its role in antibiotic tolerance in *Mycobacterium smegmatis* and identified its potential role in isoniazid resistance. We observed that the expression of *Rv1255c* upregulates the expression of different drug-resistance-related genes, chiefly those of drug efflux pumps, in *M. smegmatis*. Unexpectedly, expression of *Rv1255c* also resulted in the upregulation of *katG3* (*MSMEG_3729*), which is an isoform of *M. tuberculosis* *katG* (*Rv1908c*), the activator of the prodrug isoniazid. Reduction in intrabacterial isoniazid levels was also observed in *M. smegmatis* expressing *Rv1255c*.

Materials and methods

Bacterial strains, growth conditions and plasmids

M. tuberculosis H37Rv was cultured in Middlebrook 7H9 (BD DifcoTM) medium supplemented with 10% albumin-dextrose-catalase. For cultivation of *Mycobacterium* on plate Middlebrook 7H10 agar medium supplemented with 0.05% Tween 80, 0.5% glycerol and 10% oleate-albumin-dextrose-catalase supplement (Becton Dickinson, Franklin Lakes, NJ, USA) was used. Overexpression studies in *M. tuberculosis* was carried out with the pBEN vector system (kind gift from Prof. Lalitha Ramakrishnan, University of Cambridge, UK). All chemicals were purchased from Sigma-Aldrich (St. Louis, MO, USA) or USB (Cleveland, OH, USA). Taq DNA polymerase and primers for PCR were obtained from Sigma-Aldrich. Restriction enzymes were acquired from New England Biolabs, Ipswich, MA, USA. DNA sequencing was carried out in Applied Biosystems 3500xl Genetic Analyzer.

Stress conditions

Log phase *M. tuberculosis* was subjected to different stress conditions as follows: For heat shock, *M. tuberculosis* culture was incubated at 45 °C for 30 min. For oxidative stress induction, 2.5 mM of hydrogen peroxide (H_2O_2) was added to the *M. tuberculosis* culture and incubated at 37 °C for 40 min. For NO stress, 5 mM of diethylenetriamine/NO adduct was added, and the bacterial culture was incubated at 37 °C for 40 min. The nutrient starvation stress condition was generated by stringent washing of the log phase *M. tuberculosis* culture in phosphate-buffered saline (PBS) to remove all media components and subsequent resuspension in the same and incubation at 37 °C for 24 h. Acidic stress was induced by adding HCl to the medium to pH 4.5 and incubating at 37 °C for 24 h. For antibiotic stress, log phase cells were treated with 10 times the minimum inhibitory concentration (MIC) observed for each drug. The concentrations of drugs used were: Isoniazid—500 μg ml^{-1}, streptomycin—2 μg ml^{-1}, ethambutol—4 μg ml^{-1}, rifampicin—1.2 μg ml^{-1}. Cells were incubated with drugs for 4 h.

RNA isolation, complementary DNA synthesis and quantitative RT-PCR

RNA isolation from *M. tuberculosis* was carried out as it was performed by Gopinath et al. [7] using Trizol (Sigma-Aldrich) reagent. RNA was resuspended in nuclease-free water and treated with DNaseI (Sigma-Aldrich) at room temperature for 30 min. Complementary DNA was synthesised using a Reverse Transcriptase Core kit (Promega) according to the manufacturer's protocol. Quantitative RT-PCR reaction was performed using SYBR Green Supermix kit (Bio-Rad). The cycling conditions were as follows: an initial denaturation of 94 °C for 5 min; 35 cycles of denaturation at 94 °C for 30 s, primer annealing at 59 °C for 30 s, extension at 72 °C for 40 s, followed by a melt curve analysis. Relative gene expressions were studied using the DDCt method, with sigma factor A (*sigA*) as the housekeeping control gene.

Resazurin microtiter assay (REMA)

REMA was performed as detailed by Martin et al. [24]. The optical density of bacterial cultures was adjusted to 0.2 at 600 nm. The diluted culture (100 µl) was transferred to the wells of a 96-well microtiter plate (Nunclon; Thermo Fisher Scientific, Denmark) containing 100 µl of serially diluted stock solutions (50 µg ml^{-1}) of isoniazid, rifampicin, ethambutol, pyrazinamide and acriflavine. In the case of isoniazid for wildtype *M. smegmatis*, the stock concentration was 500 µg ml^{-1}. The plates were incubated for 24 h for *M. smegmatis* with respective controls. After incubation, MIC was determined by adding resazurin (30 µl of 0.02% aqueous solution). The blue colour of the culture indicated inhibition of bacterial growth, whereas the pink colour indicated viable bacteria. The experiment was repeated thrice.

Ethidium bromide assay

Ethidium bromide (EtBr) assay was carried out as explained by Pal et al. [25] using a log phase culture of OD 0.6. The culture was centrifuged at 5000 *g* for 10 min and was washed thrice with PBS. The washed *M. smegmatis* cultures with appropriate controls were made up to 0.3 OD in PBS, and the drug was added. EtBr was added to a final concentration of 0.5 µg ml^{-1}, and the fluorescence was monitored with excitation at 520 nm and emission at 590 nm.

Generation of *M. smegmatis* constitutively expressing Rv1255c

For constitutive expression of *Rv1255c*, ORF was amplified from H37Rv genomic DNA using specific set of primers (pBEN1255c FP-5'gatcggatccatggactacaaagacgatgacgacaaggcgggtaccgactggctgtccg 3', pBEN1255c RP- 5'aagcttt-cactcgggtccagggt 3'). PCR conditions: Initial denaturation at 95 °C for 30 s followed by the annealing of gene-specific primers (50 pmol) at 55 °C for 30 s, elongation at 72 °C for 45 s and 35 cycles. FLAG-tagged PCR product was later cloned into the MCS of pBEN plasmid [26, 27] using BamH1 and Hind III, which are located downstream of *hsp65* promoter. In-frame reading of FLAG-*Rv1255c* fusion were further verified by sequencing as well. The resultant plasmid was later transformed into electro-competent *M. smegmatis* and kanamycin-resistant transformants were selected by plating them on LB-tween-80 plate containing kanamycin (50 µg ml^{-1}) (Supplementary Fig. 1).

Homology modelling and analysis of protein structure

M. tuberculosis wildtype Kat G protein was obtained from RCSB Protein Data Base (https://www.rcsb.org). *M. smegmatis* KatG1 and KatG3 alpha fold homology models were obtained from the Uniport database. The proteins were visualised and compared using preexisting tools in Chimera software (Multi align Viewer, Matchmaker, Match → Align) as described by Meng et al. [28].

Quantification of intrabacterial isoniazid

M. smegmatis grown to mid-log phase (OD = 0.6) in 7H9 medium was treated with isoniazid (50 µg ml^{-1}) for 24 h. Bacterial cells were harvested and washed with a precooled (−80 °C) mixture of CH3CN/CH3OH/water (2:2:1) and subjected to bead beating (Mini-beadbeater, Biospec products) for cell lysis. The metabolite extract in organic solvent was collected after filtration. The sample was normalised based on protein concentration determined using the Pierce BCA protein assay. For metabolite analysis, the solvent was removed, and the sample was concentrated using a vacuum concentrator (Thermo Scientific). Vanquish UHPLC system (Thermo Scientific) with the Xcalibur acquisition software was used to isolate isoniazid, employing a reverse-phase ACQUITY UPLC BEH HILIC column (Waters) and isocratic flow conditions with 5% methanol in 0.1% formic acid. The LC-MS spectra of the metabolites were acquired using a Triple Quadrupole Mass Spectrometer (Thermo Fisher Scientific). The identity was confirmed by comparing it with the mass spectrum of pure isoniazid. The signal intensity of isoniazid was quantified by constructing a standard curve using independently prepared isoniazid standards.

Statistical analysis

Data were analysed and plotted in Origin Pro 2015 version, and Student's *t*-test was performed for statistical analysis of the data.

Results

Expression of Rv1255c is upregulated in *M. tuberculosis* under various stress conditions

Since *Rv1255c* is annotated as a hypothetical gene with a putative transcriptional regulatory activity in *M. tuberculosis* database, we began by checking the transcriptional levels of *Rv1255c* under various environmental stresses which mimic the conditions in vivo. We subjected the bacterium to different conditions such as heat, reactive oxygen species (ROS), reactive nitrogen species (NOS), nutrient starvation, and acidic pH, and studied the expression of *Rv1255c* by RT-PCR. We found that the *Rv1255c* was highly upregulated during heat stress (Fig. 1).

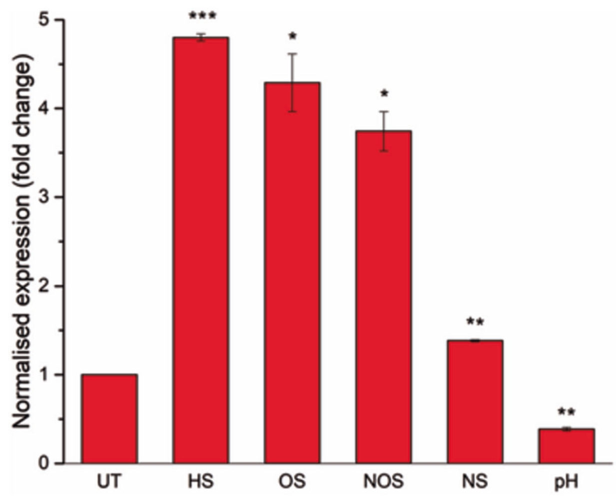

Fig. 1 Quantitative RT-PCR of *Rv1255c* in *M. tuberculosis* during stress. RNA was isolated from *M. tuberculosis* cultures untreated (UT), heat stress (HS), H_2O_2 (OS), $NaNO_2$ (NOS), nutrient starvation (NS) and low pH 4.5 (pH). The expression of the gene is repressed in normal conditions and de-repressed under stress conditions. The upregulation of *Rv1255c* was observed to be significantly higher under heat stress, oxidative, nitric oxide stresses and slightly during nutrient starvation. The expression was downregulated during pH stress (Student's *t*-test, $*p < 0.005$, $**p < 0.0005$, $***p < 0.00005$)

Likewise, upon treatment of *M. tuberculosis* with H_2O_2 to replicate the stress due to ROS, we observed significant upregulation in *Rv1255c* expression. A similar trend was observed during nitrous oxide stress as well. The expression of *Rv1255c* increased around four-fold in comparison with the standard conditions after exposure to low concentrations of NO. A relatively small, 1.4-fold increase was observed during nutrient starvation. However, downregulation of the gene was observed under acidic conditions.

Rv1255c expression causes drug tolerance to isoniazid, ethambutol, and streptomycin in *M. smegmatis*

In silico studies predicted Rv1255c as a TFTR, a class of proteins well known for its pivotal role in drug tolerance across various bacterial species [29, 30]. We tried to unravel the potential role of *M. tuberculosis* TFTR class homologue in drug resistance. As multiple attempts failed to generate a clean knock out of *Rv1255c* in *M. tuberculosis*, we used an alternative approach in which we ectopically expressed FLAG-Rv1255c fusion protein in *M. smegmatis* which is considered as a surrogate model of *M. tuberculosis*. The expression of FLAG-Rv1255c fusion protein in *M. smegmatis* was confirmed by western bloting (Supplementary Fig. 1). Using this strain, the effect of *Rv1255c* in antibiotic tolerance was studied using the rapid colourimetric REMA conducted against isoniazid, ethambutol, streptomycin, rifampicin, acriflavine and pyrazinamide. For a better comparison, we treated *M. smegmatis* harbouring an empty vector with the same antibiotics. The recombinant strain exhibited more tolerance to isoniazid (MIC >50 $\mu g\,ml^{-1}$) in comparison with the vector control, which had lesser tolerance (MIC 25 $\mu g\,ml^{-1}$). MIC of wildtype *M. smegmatis* to isoniazid was estimated to be 31.25 $\mu g\,ml^{-1}$. The REMA of *M. smegmatis* strains treated with serially diluted antibiotics (Fig. 2a) showed a similar increase in MIC against ethambutol and streptomycin (Table 1). Interestingly, no significant change was found in the MIC of acriflavine, pyrazinamide and rifampicin.

Expression of *Rv1255c* during antibiotic treatment was confirmed with quantitative RT-PCR (Fig. 2b). Higher expression of *Rv1255c* was indeed seen in the *M. smegmatis* constitutively expressing *Rv1255c*, and thus it was confirmed that the difference in MIC observed in REMA was due to the expression of *Rv1255c*, suggesting that Rv1255c increases the drug tolerance of *M. smegmatis* constitutively expressing it.

Strains constitutively expressing Rv1255c have higher efflux activity

Since many of the antibiotic resistance or tolerance mechanisms are due to the efflux system, which can pump the drugs out, we employed a fluorescence assay using EtBr to monitor efflux activities in the bacteria. EtBr assay was performed in recombinant *M. smegmatis* expressing *Rv1255c* under various antibiotic stress conditions. This strain of *M. smegmatis* showed increased fluorescence emission caused by increased efflux activity than that by the vector control (Fig. 3a–d). Higher efflux activity was observed in the strain constitutively expressing *Rv1255c* irrespective of the drug treatment, suggesting that the TFTR *Rv1255c* raises the basal efflux activity in *M. smegmatis* resulting in antibiotic tolerance.

Furthermore, we examined the expression of the genes in *M. smegmatis* orthologous to the central efflux pumps and genes associated with antibiotic tolerance in *M. tuberculosis* (Supplementary Table 1). Consistent with the above observations, we observed that efflux pumps *MSMEG_3118 (drrB), MSMEG_3763 (drrB), MSMEG_5779* (*pstB*), *MSMEG_5124 (fadH)* are upregulated (Fig. 3e) in the strain constitutively expressing *Rv1255c* without antibiotic treatment. In addition, isoniazid resistance-associated *katG* genes, *MSMEG_6384 (katG1) and MSMEG_3729 (katG3)* are also significantly upregulated under the influence of *Rv1255c* while *MSMEG_3461 (katG2)* was downregulated (Fig. 4a).

Reduced intrabacterial isoniazid concentration

To validate the increased expression levels of efflux pumps in *M. smegmatis* expressing *Rv1255c*, we performed an intrabacterial drug quantification assay. Intrabacterial

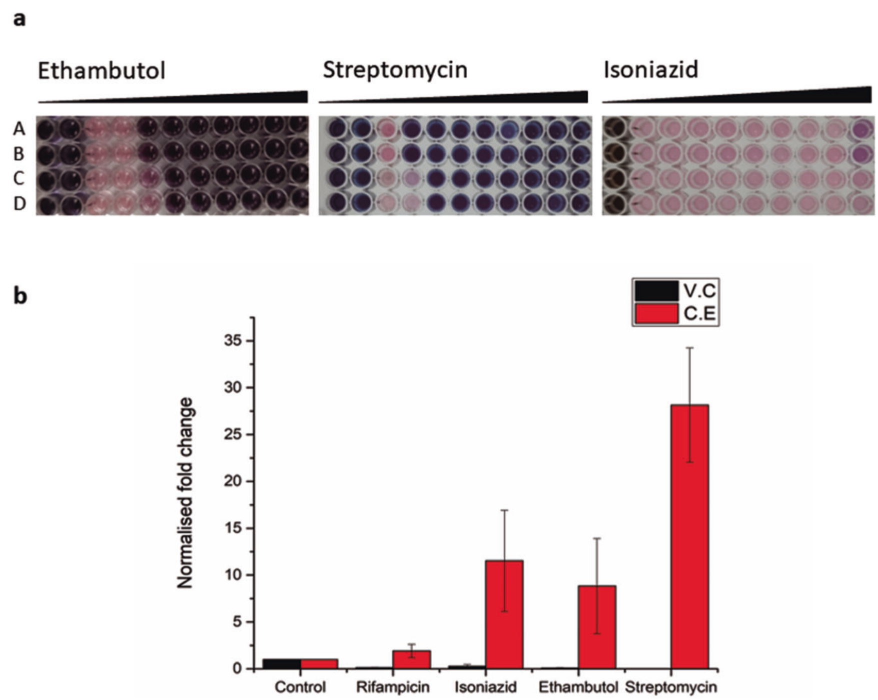

Fig. 2 **a** REMA of *M. smegmatis*. Wells A and B are *M. smegmatis* with the vector only, and wells C and D are *M. smegmatis* constitutively expressing *Rv1255c*. MIC of *M. smegmatis* constitutively expressing *Rv1255c* is increased to 50 μg ml^{-1} compared to the MIC of vector control 25 μg ml^{-1} in isoniazid-treated bacteria. MIC of streptomycin increased from 0.195 to 0.390 μg ml^{-1} and MIC of ethambutol increased from 0.390 to 0.780 μg ml^{-1}. **b** Expression of *Rv1255c* during antibiotic treatment was confirmed in *M. smegmatis* constitutively expressing *Rv1255c* (Student's *t*-test, $p < 0.05$)

Table 1 Minimum inhibitory concentration (MIC) of *M. smegmatis* constitutively expressing *Rv1255c* to antibiotics

Antibiotic	MIC of wildtype (μg ml^{-1})	MIC of Vector control (μg ml^{-1})	MIC of Rv1255c constitutive expression (μg ml^{-1})
Isoniazid	31.25	25	50
Streptomycin	0.195	0.195	0.39
Ethambutol	0.195	0.39	0.781
Rifampicin	3.125	3.125	3.125
Pyrazinamide	0.0965	<0.195	<0.195
Acriflavine	3.125	6.25	6.25

isoniazid was extracted, purified, and quantified using HPLC coupled with mass spectrometry. After 24 h of treatment with isoniazid, the intracellular isoniazid levels in three isolates of *M. smegmatis* expressing *Rv1255c* were found to be consistently lower (0.243, 0.212 and 0.206 μM) than the vector control (0.36344 μM) (Fig. 5). These findings suggest that the observed increase in expression levels of efflux pumps in *M. smegmatis* expressing *Rv1255c* could potentially account for the reduction in intracellular isoniazid concentrations.

Isoniazid-resistant substitutions in Kat G3

To understand the relation between isoniazid tolerance and the overexpressed *katG*s of *M. smegmatis*, the structure of KatG (Rv1908) was compared with the alpha fold homology models of KatG1 (MSMEG_6384), KatG2 (MSMEG_3461) and KatG3 (MSMEG_3729). The percentage identity of KatG1 with Rv1908 is 70%, Kat G2 is 64%, whereas that of KatG3 with Rv1908 is 59%. The overall structure of the three proteins is very similar. Notably, there are substitutions in the positions of high-confidence mutations in both M. *smegmatis* KatG proteins (Supplementary Table 2). Though the very crucial 315S [31] is conserved in both proteins, KatG3 has S140N, G279T, G593A and S457A substitutions. KatG1 has an S457A substitution.

Discussion

Currently, TB is one of the most challenging health problems due to the more prolonged drug treatment regime and rising numbers of MDR and XDR cases. However, this pathogen's success is also attributable to its ability to stay dormant in the body under high antimicrobial stress, and to remain tolerant to anti-tubercular drugs, and reactivate only when the host's immune system deteriorates [32]. Due to this reason, latent tuberculosis infection acts as a reservoir for future infections, and thus, it needs to be tackled with equal importance as that of active TB to eradicate the disease.

We have demonstrated that *Rv1255c* is overexpressed during heat, oxidative, nitrosative, and nutrient stresses. Expression of *Rv1255c* was also investigated in previously published studies and databases. It was found that *Rv1255c*

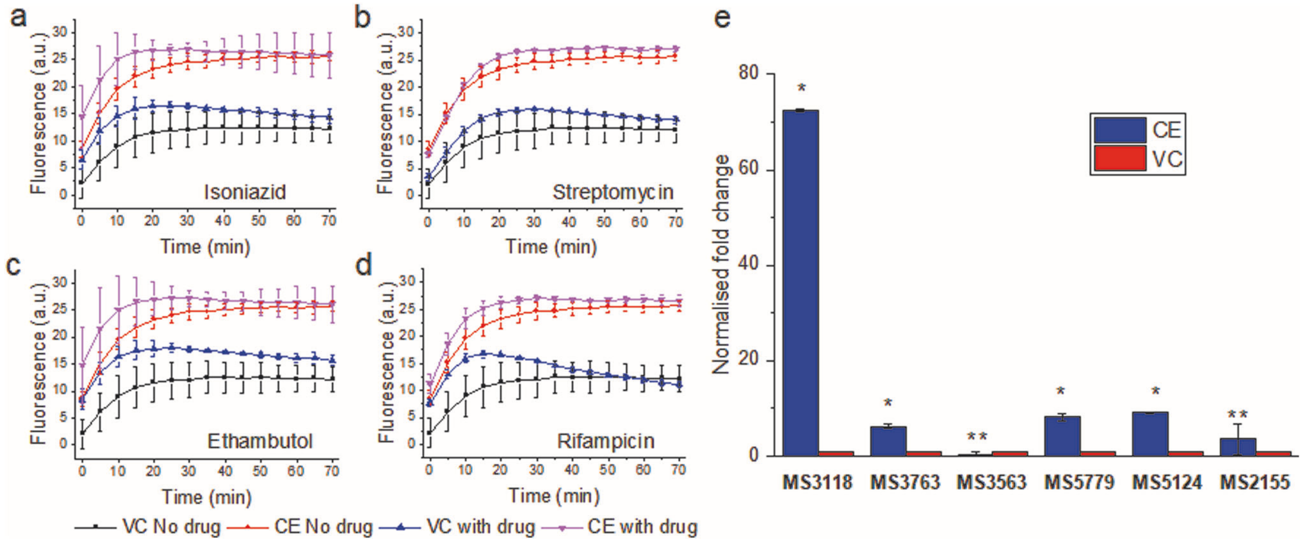

Fig. 3 Ethidium bromide assay conducted in *M. smegmatis* under **a** isoniazid, **b** streptomycin, **c** ethambutol, **d** rifampicin treatment. *M. smegmatis* expressing *Rv1255c* (CE) has increased efflux activity than that of the vector control (VC). **e** Relative expression of drug resistance-associated genes in *M. smegmatis* expressing *Rv1255c* (Student's *t*-test, $*p < 0.005$, $**p < 0.5$)

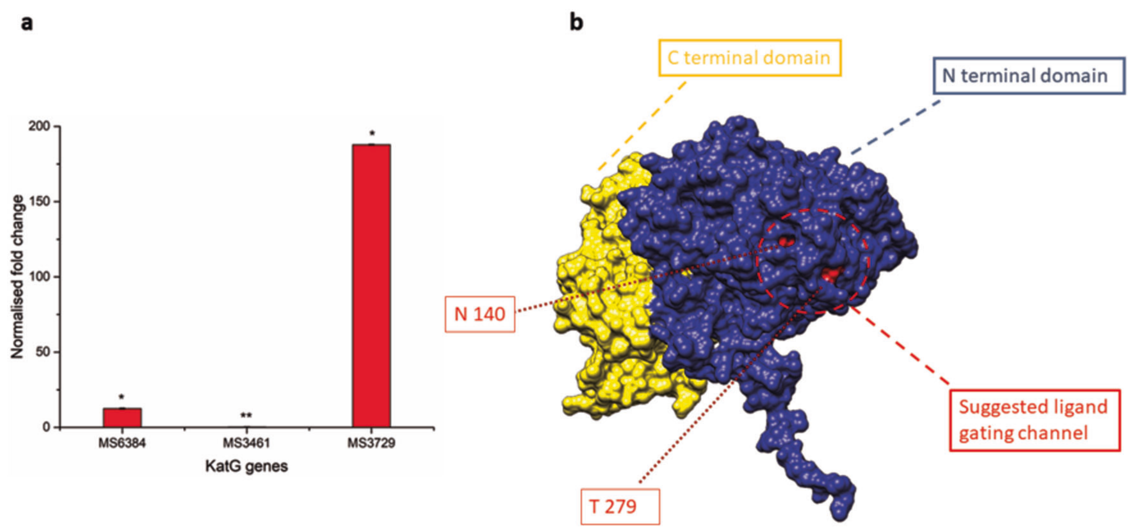

Fig. 4 **a** Relative expression of *katG* isoforms in *M. smegmatis* expressing *Rv1255c* (Student's *t*-test, $*p < 0.005$, $**p < 0.05$). **b** Kat G3 homology model showing isoniazid-resistant substitutions at its suggested isoniazid gating channel indicated in red as S140N, G279T

is overexpressed as an enduring hypoxic response gene during the later stages of hypoxia [21]. It is deduced that *Rv1255c* is overexpressed in *M. tuberculosis* during some of the major stress conditions that cause dormancy. Moreover, Sun et al. showed that Rv3334, a master regulator downstream of DosR regulon, positively regulates *Rv1255c* and other enduring hypoxic response genes [22, 33]. This enduring hypoxic gene, *Rv1255c* causes further activation and repression of many downstream genes and results in physiological changes such as antibiotic tolerance. Our results showed that constitutive expression of *Rv1255c* enhanced drug tolerance to isoniazid, ethambutol, and streptomycin in *M. smegmatis*.

Widely accepted mechanisms of drug resistance include reduction in the uptake of a drug, mutation of a drug target, deactivation of a drug, and active efflux of a drug. Recent studies have indicated that several mycobacterial efflux pumps are coupled with drug resistance, yet the regulatory mechanisms of drug efflux are unknown. Some transcriptional regulators were identified as activators and repressors of these drug transport proteins. For example, EthR in *M. tuberculosis* is a repressor belonging to TFTR, which is associated with ethionamide resistance [29]. TFTRs take part in disparate physiological functions as positive and negative regulators and their most important function can be considered as regulating antibiotic resistance. Similarly, it

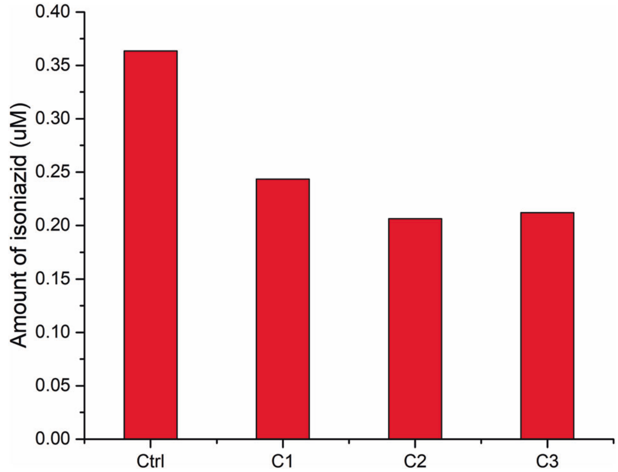

Fig. 5 Intrabacterial drug quantification: amount of intrabacterial isoniazid in the three recombinant *M. smegmatis* expressing *Rv1255c* (C1, C2, C3) compared to the vector-only control (ctrl)

was recently shown in *M. smegmatis* that the TFTR MSMEG_4022 positively regulates the activation of transport-related genes and causes rifampicin resistance in the bacteria [17].

We showed that the basal efflux activity of the bacteria is increased by the constitutive expression of Rv1255c, a TFTR, by enhancing the expression of efflux genes. We observed that the expression of *Rv1255c* increases the overall efflux activity of the bacterium and upregulates the expression of different drug-resistance-related genes such as *MSMEG_3118*, *MSMEG_3763,* *MSMEG_5779* and *MSMEG_5124* in *M. smegmatis*. It should be noted that *MSMEG_3118* and *MSMEG_3763* are orthologous genes of *drrB* (ABC transporter), *MSMEG_5779* is that of *pstB* (ABC transporter) and *MSMEG_5124* is that of *fadH* (2,4-dienoyl-coA reductase) which are correlated to isoniazid resistance and multidrug resistance in *M. tuberculosis* [14]. To validate this correlation, intrabacterial isoniazid content in *M. smegmatis* constitutively expressing *Rv1255c* was estimated, and the results showed a reduction in the isoniazid levels in the strains overexpressing efflux pumps. The findings strongly indicate that the observed upregulation of efflux pump expression in *M. smegmatis* constitutively expressing *Rv1255c* is likely responsible for the reduction in intracellular isoniazid concentration, which explains the change in isoniazid's MIC. This moderate change can be attributed to a consistent expression of *Rv1255c* rather than its overexpression, as demonstrated by western blot analysis of the recombinant strain (Supplementary Fig. 1).

Furthermore, isoniazid is a prodrug activated by the intracellular KatG enzyme of *M. tuberculosis* [34]. Mutations in katG enzyme are known to cause isoniazid resistance [31]. In contrast *to M. tuberculosis*, which has only one KatG enzyme, *M. smegmatis* possesses three KatG enzymes. In our study, we observed that Rv1255c upregulates *katG1* and *katG3*, and downregulates *katG2*. This is interesting because KatG, the activator of isoniazid, is expected to influence antibiotic treatment positively. But here, counterintuitively, it makes the bacteria tolerant to isoniazid. A previous study related to *furA* gene has also observed that isoniazid-resistant *M. smegmatis* strains show *katG3* upregulation and *katG1/katG2* downregulation [35]. Thus, it can be inferred that isoniazid resistance is likely caused by the downregulation of *katG2* or the upregulation of *katG3*, rather than by the levels of *katG1*. According to Barozi et al., there are 11 high-confidence mutations of KatG that cause conformational changes in the enzyme rendering it resistant to isoniazid [36]. Therefore, we made a comparison of this isoform, KatG3, with KatG of *M. tuberculosis*, and found that KatG3 has substitutions at the vital positions that are required for the activation of isoniazid. This could mean that highly upregulated KatG3 could be inhibiting the activation of the prodrug isoniazid, which is leading to isoniazid tolerance of the bacterium. Hence, understanding the role of transcriptional regulators like *Rv1255c*, will help to manage the problem of drug tolerance during dormancy. However, replicating the study in pathogenic *M. tuberculosis* and assessment of KatG3 resistance to isoniazid activation would provide additional insights into *Rv1255c*-mediated drug tolerance in *M. tuberculosis*.

Thus, our study indicates an interesting mechanism by which dormant *M. tuberculosis* could modulate drug-tolerance-related genes by inducing multiple transcriptional regulatory proteins such as *Rv1255c* in response to stress stimuli. According to our study, Rv1255c in *M. smegmatis* enhances its tolerance to isoniazid through the regulation of both the drug efflux system and intrabacterial isoniazid levels. In addition, in our effort to understand the mechanism of drug tolerance caused by Rv1255c, we found that *M. smegmatis* KatG3 protein harbours multiple amino acid substitutions associated with isoniazid resistance.

Acknowledgements JGR thanks the Department of Science and Technology, Government of India, for the INSPIRE fellowship. KK thanks the Department of Biotechnology, Government of India for the Ramalingaswami fellowship. RAK is grateful to the Department of Biotechnology, Government of India for financial support.

Compliance with ethical standards

Conflict of interest The authors declare no competing interests.

References

1. Bates JH, Stead WW. The history of tuberculosis as a global epidemic. Med Clin North Am. 1993;77:1205–17.

2. WHO. Global Tuberculosis Report 2020 – World. ReliefWeb. 2020;1–232.
3. World Health Organization. Global Tuberculosis Report 2021. Geneva: ©World Health Organization; 2021. p 57.
4. Gengenbacher M, Kaufmann SHE. Mycobacterium tuberculosis: success through dormancy. FEMS Microbiol Rev. 2012;36:514–32.
5. Trutneva KA, Shleeva MO, Demina GR, Vostroknutova GN, Kaprelyans AS. One-year old dormant, "non-culturable" *Mycobacterium tuberculosis* preserves significantly diverse protein profile. Front Cell Infect Microbiol. 2020;10:26.
6. Wayne LG, Sohaskey CD. Nonreplicating persistence of *Mycobacterium tuberculosis*. Annu Rev Microbiol. 2001;55:139–63.
7. Gopinath V, Raghunandanan S, Gomez RL, Jose L, Surendran A, Ramachandran R, et al. Profiling the proteome of *Mycobacterium tuberculosis* during dormancy and reactivation. Mol Cell Proteom. 2015;14:2160–76.
8. Wayne LG, Hayes LG. An in vitro model for sequential study of shiftdown of *Mycobacterium tuberculosis* through two stages of nonreplicating persistence. Infect Immun. 1996;64:2062–9.
9. Ojha AK, Baughn AD, Sambandan D, Hsu T, Trivelli X, Guerardel Y, et al. Growth of *Mycobacterium tuberculosis* biofilms containing free mycolic acids and harbouring drug-tolerant bacteria. Mol Microbiol. 2008;69:164–74.
10. Flentie K, Harrison GA, Tükenmez H, Livny J, Good JAD, Sarkar S, et al. Chemical disarming of isoniazid resistance in *Mycobacterium tuberculosis*. Proc Natl Acad Sci USA. 2019;116:10510–7.
11. Voskuil MI, Bartek IL, Visconti K, Schoolnik GK. The response of *Mycobacterium tuberculosis* to reactive oxygen and nitrogen species. Front Microbiol. 2011;2:105.
12. Deb C, Lee C-M, Dubey VS, Daniel J, Abomoelak B, Sirakova TD, et al. A novel in vitro multiple-stress dormancy model for *Mycobacterium tuberculosis* generates a lipid-loaded, drug-tolerant, dormant pathogen. PLoS One. 2009;4:e6077.
13. Shastri MD, Shukla SD, Chong WC, Dua K, Peterson GM, Patel RP, et al. Role of oxidative stress in the pathology and management of human tuberculosis. Oxid Med Cell Longev. 2018;2018:7695364.
14. Li G, Zhang J, Guo Q, Jiang Y, Wei J, Zhao LL, et al. Efflux pump gene expression in multidrug-resistant *Mycobacterium tuberculosis* clinical isolates. PLoS One. 2015;10:e0119013.
15. Bhandol H, Alindogan J, De Guzman A, Lim R. Review: Structure and transcriptional regulation of the Acr efflux pumps and their role in antibiotic resistance in *Escherichia coli*. Undergrad J Exp Microbiol Immunol. 2020;6:1–16.
16. Liu J, Shi W, Zhang S, Hao X, Maslov DA, Shur KV, et al. Mutations in efflux pump Rv1258c (Tap) cause resistance to pyrazinamide, isoniazid, and streptomycin in *M. tuberculosis*. Front Microbiol. 2019;10:216.
17. Liu H, Yang M, He Z-G. Novel TetR family transcriptional factor regulates expression of multiple transport-related genes and affects rifampicin resistance in *Mycobacterium smegmatis*. Sci Rep. 2016;6:27489.
18. Remm S, Earp JC, Dick T, Dartois V, Seeger MA. Critical discussion on drug efflux in *Mycobacterium tuberculosis*. FEMS Microbiol Rev. 2022;46:fuab050.
19. Balhana RJC, Singla A, Sikder MH, Withers M, Kendall SL. Global analyses of TetR family transcriptional regulators in mycobacteria indicates conservation across species and diversity in regulated functions. BMC Genomics. 2015;16:1–12.
20. Luo L, Zhu L, Yue J, Liu J, Liu G, Zhang X, et al. Antigens Rv0310c and Rv1255c are promising novel biomarkers for the diagnosis of *Mycobacterium tuberculosis* infection. Emerg Microbes Infect. 2017;6:1–8.
21. Rustad TR, Harrell MI, Liao R, Sherman DR. The enduring hypoxic response of *Mycobacterium tuberculosis*. PLoS One. 2008;3:e1502.
22. Sun X, Zhang L, Jiang J, Ng M, Cui Z, Mai J, et al. Transcription factors Rv0081 and Rv3334 connect the early and the enduring hypoxic response of *Mycobacterium tuberculosis*. Virulence. 2018;9:1468.
23. Galagan JE, Minch K, Peterson M, Lyubetskaya A, Azizi E, Sweet L, et al. The *Mycobacterium tuberculosis* regulatory network and hypoxia. Nature. 2013;499:178–83.
24. Martin A, Camacho M, Portaels F, Palomino JC. Resazurin microtiter assay plate testing of *Mycobacterium tuberculosis* susceptibilities to second-line drugs: rapid, simple, and inexpensive method. Antimicrob Agents Chemother. 2003;47:3616.
25. Pal S, Misra A, Banerjee S, Dam B. Adaptation of ethidium bromide fluorescence assay to monitor activity of efflux pumps in bacterial pure cultures or mixed population from environmental samples. J King Saud Univ Sci. 2020;32:939–45.
26. Valdivia RH, Hromockyj AE, Monack D, Ramakrishnan L, Falkow S. Applications for green fluorescent protein (GFP) in the study of host-pathogen interactions. Gene. 1996;173:47–52.
27. Jose L, Ramachandran R, Bhagavat R, Gomez RL, Chandran A, Raghunandanan S, et al. Hypothetical protein Rv3423.1 of *Mycobacterium tuberculosis* is a histone acetyltransferase. FEBS J. 2016;283:265–81.
28. Meng EC, Pettersen EF, Couch GS, Huang CC, Ferrin TE. Tools for integrated sequence-structure analysis with UCSF Chimera. BMC Bioinforma. 2006;7:1–10.
29. Engohang-Ndong J, Baillat D, Aumercier M, Bellefontaine F, Besra GS, Locht C, et al. EthR, a repressor of the TetR/CamR family implicated in ethionamide resistance in mycobacteria, octamerizes cooperatively on its operator. Mol Microbiol. 2004;51:175–88.
30. Ramos JL, Martínez-Bueno M, Molina-Henares AJ, Terán W, Watanabe K, Zhang X, et al. The TetR family of transcriptional repressors. Microbiol Mol Biol Rev. 2005;69:326–56.
31. Bollela VR, Namburete EI, Feliciano CS, Macheque D, Harrison LH, Caminero JA. Detection of katG and inhA mutations to guide isoniazid and ethionamide use for drug-resistant tuberculosis. Int J Tuberc Lung Dis. 2016;20:1099.
32. Ai JW, Ruan QL, Liu QH, Zhang WH. Updates on the risk factors for latent tuberculosis reactivation and their managements. Emerg Microbes Infect. 2016;5:e10.
33. Gomez RL, Jose L, Ramachandran R, Raghunandanan S, Muralikrishnan B, Johnson JB, et al. The multiple stress responsive transcriptional regulator Rv3334 of *Mycobacterium tuberculosis* is an autorepressor and a positive regulator of kstR. FEBS J. 2016;283:3056–71.
34. Ramaswamy S, Musser JM. Molecular genetic basis of antimicrobial agent resistance in *Mycobacterium tuberculosis*: 1998 update. Tube Lung Dis. 1998;79:3–29.
35. Gao CH, Wei WP, Tao HL, Cai LK, Jia WZ, Hu L, et al. Crosstalk between the three furA orthologs in *Mycobacterium smegmatis* and the contribution to isoniazid resistance. J Biochem. 2019;166:237–43.
36. Barozi V, Musyoka TM, Sheik Amamuddy O, Tastan Bishop Ö. Deciphering isoniazid drug resistance mechanisms on dimeric *Mycobacterium tuberculosis* KatG via post-molecular dynamics analyses including combined dynamic residue network metrics. ACS Omega. 2022;7:13313–32.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.